

Practice Questions for Weak Duality

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1 Questions

1. In the context of weak duality, what is the relationship between the objective function values of the primal and dual problems?
 - a. The primal objective function value is always greater than or equal to the dual objective function value.
 - b. The primal objective function value is always less than or equal to the dual objective function value.
 - c. The primal and dual objective function values are always equal.
 - d. There is no relationship between the primal and dual objective function values.
 - e. The relationship between the primal and dual objective function values depends on the specific problem instance.
2. The Klee-Minty problem demonstrates that the simplex method can exhibit exponential time complexity in the worst case. What is the fundamental characteristic of the Klee-Minty problem that leads to this behavior?
 - a. The presence of many constraints.
 - b. The presence of many variables.
 - c. The highly skewed geometry of its feasible region.
 - d. The use of a specific pivot rule.
 - e. The use of a particular objective function.
3. What is the significance of weak duality in linear programming?
 - a. It guarantees the existence of an optimal solution.
 - b. It provides a bound on the optimal objective function value.
 - c. It ensures the uniqueness of the optimal solution.
 - d. It determines the choice of the pivot rule in the simplex method.
 - e. It is only relevant for problems with equality constraints.
4. Empirical studies suggest that the average-case performance of the simplex method is polynomial. What is the practical implication of this finding?

- a. The simplex method is guaranteed to solve any linear program in polynomial time.
 - b. The simplex method is always faster than interior-point methods.
 - c. The simplex method is practically efficient for a wide range of real-world problems.
 - d. The worst-case exponential complexity of the simplex method is irrelevant.
 - e. All pivot rules for the simplex method have polynomial average-case complexity.
5. In the context of weak duality, if Z represents the objective function value of the primal problem and W represents the objective function value of the dual problem, which inequality always holds?
- a. $Z \geq W$
 - b. $Z \leq W$
 - c. $Z = W$
 - d. $Z > W$
 - e. $Z < W$
6. Suppose the primal problem has an unbounded objective function. What can we conclude about the dual problem based on weak duality?
- a. The dual problem is also unbounded.
 - b. The dual problem is infeasible.
 - c. The dual problem has a finite optimal solution.
 - d. The dual problem is unbounded or infeasible.
 - e. There is no relationship between the primal and dual problems.
7. Consider a primal linear programming problem and its corresponding dual problem. If the primal problem is infeasible, what can be concluded about the dual problem based on weak duality?
- a. The dual problem is also infeasible.
 - b. The dual problem is unbounded.
 - c. The dual problem has a finite optimal solution.
 - d. The dual problem is either unbounded or infeasible.
 - e. No conclusion can be drawn about the dual problem.
8. The primal problem is: Maximize $Z = 2x_1 + 3x_2$ subject to $x_1 + x_2 \leq 5$, $x_1 \leq 3$, $x_2 \leq 4$, $x_1, x_2 \geq 0$. A feasible solution to the primal problem is $x_1 = 2, x_2 = 2$. The objective function value is $Z = 10$. Using weak duality, what can you conclude about the objective function value W of the dual problem?
- a. $W \leq 10$
 - b. $W \geq 10$
 - c. $W = 10$
 - d. $W < 10$

- e. No conclusion can be made about W
9. In the context of weak duality, if Z represents the objective function value of the primal problem and W represents the objective function value of the dual problem, what inequality always holds between Z and W for a maximization primal problem?
- a. $Z \geq W$
 - b. $Z \leq W$
 - c. $Z = W$
 - d. $Z > W$
 - e. $Z < W$
10. In the context of weak duality, if Z represents the objective function value of the primal maximization problem and W represents the objective function value of the dual minimization problem, which inequality always holds?
- a. $Z \leq W$
 - b. $Z \geq W$
 - c. $Z = W$
 - d. $Z > W$
 - e. $Z < W$

2 Answers

1. In the context of weak duality, what is the relationship between the objective function values of the primal and dual problems?
 - a. This is incorrect. While it's true that the primal objective function value is greater than or equal to the dual objective function value at optimality (strong duality), this is not always the case for arbitrary feasible solutions. Weak duality only guarantees that the dual objective function value is a lower bound for the primal objective function value.
 - b. Weak duality states that the objective function value of any feasible solution to the dual problem is always less than or equal to the objective function value of any feasible solution to the primal problem. This holds true regardless of whether the primal or dual problem is a maximization or minimization problem.
 - c. This is incorrect. The primal and dual objective function values are only equal at optimality (strong duality), not for arbitrary feasible solutions. Weak duality only provides an inequality relationship.
 - d. This is incorrect. Weak duality establishes a specific relationship between the objective function values of the primal and dual problems.
 - e. This is incorrect. The relationship is consistent and defined by weak duality, regardless of the specific problem instance.
2. The Klee-Minty problem demonstrates that the simplex method can exhibit exponential time complexity in the worst case. What is the fundamental characteristic of the Klee-Minty problem that leads to this behavior?
 - a. While the number of constraints can influence the runtime, it's not the fundamental reason for the exponential behavior in the Klee-Minty problem. The problem's structure, specifically the skewed geometry, is the key factor.
 - b. Similar to the number of constraints, the number of variables contributes to the problem's size, but the skewed geometry is the crucial element causing the exponential runtime.
 - c. The Klee-Minty problem's feasible region is a highly skewed polytope. The simplex method, following certain pivot rules, can traverse a path along the boundary of this polytope that is exponentially long in the number of variables, leading to exponential time complexity.
 - d. While the choice of pivot rule affects the path taken by the simplex method, the inherent structure of the Klee-Minty problem's feasible region is the primary reason for its exponential complexity. Different pivot rules might lead to different paths, but the exponential length of the worst-case path is a property of the problem itself.
 - e. The objective function in the Klee-Minty problem is designed to interact with the skewed geometry of the feasible region to create the exponential path. However, the skewed geometry is the fundamental characteristic causing the exponential behavior.
3. What is the significance of weak duality in linear programming?
 - a. The existence of an optimal solution is not guaranteed by weak duality. It's a separate concept related to the properties of the feasible region and the objective function.
 - b. Weak duality states that the objective function value of any feasible solution to the dual problem provides a lower bound (for maximization problems) or an upper bound (for minimization problems) for the optimal objective function value of the primal problem. This is significant because it allows us to assess the quality of feasible solutions without necessarily finding the optimal solution.

- c. Weak duality does not address the uniqueness of the optimal solution. Uniqueness is a separate property that depends on the structure of the problem.
 - d. Weak duality is a theoretical concept that doesn't directly influence the choice of pivot rule in the simplex method. Pivot rules are algorithmic choices aimed at improving efficiency.
 - e. Weak duality applies to both problems with equality and inequality constraints. It's a fundamental relationship between primal and dual problems regardless of the type of constraints.
4. Empirical studies suggest that the average-case performance of the simplex method is polynomial. What is the practical implication of this finding?
- a. This is incorrect. The average-case performance being polynomial does not guarantee polynomial time for all instances. The Klee-Minty problem demonstrates that the worst-case complexity is exponential.
 - b. This is incorrect. The relative efficiency of the simplex method and interior-point methods depends on the specific problem instance. While the average-case complexity of the simplex method is polynomial, interior-point methods also have polynomial worst-case complexity.
 - c. The polynomial average-case complexity suggests that, in practice, the simplex method is efficient for most real-world linear programming problems. While it has a worst-case exponential complexity, this worst-case scenario rarely occurs in practice.
 - d. This is incorrect. While the worst-case scenario is rare, understanding the worst-case complexity is important for theoretical analysis and for identifying potential limitations.
 - e. This is incorrect. The polynomial average-case complexity is observed for some pivot rules, but not all. The Klee-Minty problem demonstrates that some pivot rules can lead to exponential time complexity in the worst case.
5. In the context of weak duality, if Z represents the objective function value of the primal problem and W represents the objective function value of the dual problem, which inequality always holds?
- a. This inequality is incorrect. While it can hold in some cases, it is not universally true for all primal-dual pairs.
 - b. Weak duality states that the objective function value of the primal problem is always less than or equal to the objective function value of the dual problem. Therefore, $Z \leq W$ always holds.
 - c. This equality only holds in the case of strong duality, where the optimal objective function values of the primal and dual problems are equal. Weak duality is a broader statement that applies even when the problems are not at their optimal solutions.
 - d. This inequality is incorrect. It is not universally true for all primal-dual pairs.
 - e. This inequality is incorrect. It is not universally true for all primal-dual pairs.
6. Suppose the primal problem has an unbounded objective function. What can we conclude about the dual problem based on weak duality?
- a. This is incorrect. If the primal is unbounded, the dual must be infeasible.
 - b. If the primal problem is unbounded, meaning its objective function can be made arbitrarily large, then the dual problem must be infeasible. This is a direct consequence of weak duality ($Z \leq W$), as an unbounded Z would contradict a finite W .
 - c. This is incorrect. If the primal is unbounded, the dual must be infeasible.

- d. While the dual could be infeasible, it cannot be unbounded. If the dual were unbounded, it would imply the primal is infeasible, which is not the case here.
 - e. Weak duality establishes a relationship between the primal and dual problems. The unboundedness of the primal directly impacts the feasibility of the dual.
7. Consider a primal linear programming problem and its corresponding dual problem. If the primal problem is infeasible, what can be concluded about the dual problem based on weak duality?
- a. This is incorrect. Infeasibility of the primal implies unboundedness of the dual.
 - b. If the primal problem is infeasible, then its objective function value Z is undefined (or can be considered $-\infty$). Weak duality ($Z \leq W$) implies that the dual problem must be unbounded ($W = \infty$) to satisfy the inequality.
 - c. This is incorrect. Infeasibility of the primal implies unboundedness of the dual.
 - d. The dual cannot be infeasible in this case. If both primal and dual were infeasible, weak duality would not provide any information.
 - e. Weak duality provides a relationship between the primal and dual problems, allowing us to draw conclusions about one based on the other.
8. The primal problem is: Maximize $Z = 2x_1 + 3x_2$ subject to $x_1 + x_2 \leq 5$, $x_1 \leq 3$, $x_2 \leq 4$, $x_1, x_2 \geq 0$. A feasible solution to the primal problem is $x_1 = 2$, $x_2 = 2$. The objective function value is $Z = 10$. Using weak duality, what can you conclude about the objective function value W of the dual problem?
- a. This is incorrect. Weak duality dictates that the dual objective function value is greater than or equal to the primal objective function value.
 - b. Weak duality states that $Z \leq W$. Since $Z = 10$ for a feasible solution, the objective function value of the dual problem must be greater than or equal to 10. Therefore, $W \geq 10$.
 - c. This is incorrect. This would only be true if strong duality holds, which is not guaranteed.
 - d. This is incorrect. Weak duality dictates that the dual objective function value is greater than or equal to the primal objective function value.
 - e. Weak duality provides a direct relationship between the primal and dual objective function values.
9. In the context of weak duality, if Z represents the objective function value of the primal problem and W represents the objective function value of the dual problem, what inequality always holds between Z and W for a maximization primal problem?
- a. Weak duality states that for a maximization primal problem, the objective function value of any feasible solution to the primal problem is less than or equal to the objective function value of any feasible solution to the dual problem. Therefore, $Z \leq W$ always holds.
 - b. This inequality would be correct for a minimization primal problem, but not for a maximization primal problem.
 - c. This equality only holds at optimality, which is the statement of strong duality, not weak duality.
 - d. This inequality is not always true; weak duality only guarantees $Z \leq W$.
 - e. This inequality is not always true; weak duality only guarantees $Z \leq W$.

10. In the context of weak duality, if Z represents the objective function value of the primal maximization problem and W represents the objective function value of the dual minimization problem, which inequality always holds?
- a. Weak duality states that the objective function value of the primal maximization problem is always less than or equal to the objective function value of the dual minimization problem. Therefore, $Z \leq W$ always holds.
 - b. This inequality is incorrect. Weak duality states that the objective function value of the primal maximization problem is always less than or equal to the objective function value of the dual minimization problem.
 - c. This equality only holds at optimality (strong duality). Weak duality only guarantees an inequality.
 - d. This inequality is incorrect. Weak duality states that the objective function value of the primal maximization problem is always less than or equal to the objective function value of the dual minimization problem.
 - e. This inequality is incorrect. Weak duality states that the objective function value of the primal maximization problem is always less than or equal to the objective function value of the dual minimization problem.